TOPIC 2 Transforming and Comparing Linear Functions

Name _____ Date _____

Reminders

Slope = $(y_2 - y_1) \div (x_2 - x_1)$ · $y = mx + b$ · point-slope: $y - y_1 = m(x - x_1)$

Parallel: same slope · Perpendicular: flip the fraction and change the sign (product −1)

Outside f(x) + k: up/down · Inside f(x − h): right h, f(x + h): left h (opposite the sign)

Horizontal: y = #, slope 0 · Vertical: x = #, slope undefined (HOY VUX)

Domain = inputs (x) · Range = outputs (y) · Zero = x-intercept

1. What are the equation and slope of the vertical line through (−7, 2)?

Tip: HOY VUX: Vertical → Undefined slope → x = #.

A. y = 2; slope is zero

B. x = −7; slope is zero

C. **x = −7; slope is undefined**

Vertical: every point has x = −7. Slope = rise ÷ 0: undefined (VUX).

2. Write the equation in slope-intercept form of the line through (−1, 0) that is parallel to $y = 5x - 2$.

> **Tip:** Parallel → same slope. Then put in the point and solve for b.

Slope = _____

0 = ___(−1) + b → b = _____

y = _____x + _____

***y* = 5*x* + 5**

Same slope 5. 0 = 5(−1) + b → b = 5.

3. $f(x) = 2x + 1$ is translated 3 units left. What is the y-intercept of the new function?

> **Tip:** Left 3 means f(x + 3). Put (x + 3) in for x.

g(x) = 2(x + 3) + 1

g(x) = 2x + _____ + 1 = 2x + _____

y-intercept = _____

7

g(x) = f(x + 3) = 2(x + 3) + 1 = 2x + 7. y-intercept 7.